

Module1

Basics of Computer graphics and Algorithms

Computer Graphics

Computer graphics is an art of drawing pictures on computer screens with the help of programming. It is the Creation, Manipulation and Storage of geometric objects (modeling) & their images (rendering) and display those images on screens or hardcopy devices.

Applications of Computer Graphics

1. Computer Aided Design (CAD)
2. Presentation Graphics
3. Computer Art
4. Entertainment (animation, games, ...)
5. Education & Training
6. Visualization (scientific & business)
7. Image Processing
8. Graphical User Interfaces

1. CAD

Computer-aided design methods are used in the design of buildings, automobiles, aircraft, watercraft, spacecraft, computers, textiles, and many other products. Objects are first displayed in a wireframe outline form that shows the overall shape and internal features of objects. Software packages provide multiwindow environments. Circuits and networks for communications, water supply, etc are constructed with repeated placement of a few graphical shapes. Animations are often used in CAD applications. Real-time animations using wireframe displays on a video monitor are useful for testing performance of a vehicle or system. Animations in virtual reality environments are used to determine how vehicle operators are affected by certain motions.

Architects use interactive graphics methods to lay out floor plans, that show the positioning of rooms, doors, windows, stairs, shelves, counters, and other building features. Many other kinds of systems and products are designed using either general CAD packages or specially developed CAD software.

2. PRESENTATION GRAPHICS

Presentation graphics is used to produce illustrations for reports or to generate 35-mm slides or transparencies for use with projectors. Presentation graphics is commonly used to summarize financial, statistical, mathematical, scientific, and economic data for research reports, managerial reports, consumer information bulletins etc. Examples of presentation graphics are bar charts, line graphs, surface graphs, pie charts, and other displays showing relationships between multiple parameters. Three-dimensional graphs can provide a more dramatic or more attractive presentation of data relationships.

Time charts and task network layouts are used in project management to schedule and monitor the progress of projects.

3. COMPUTER ART

Computer graphics methods are widely used in both fine art and commercial art applications. Artists use a variety of computer methods, including special-purpose hardware, artist's paintbrush (Lumena), paint packages (Pixelpaint and Superpaint), specially developed software, symbolic mathematics packages (Mathematics), CAD packages, desktop publishing software, and animation packages that provide facilities

for designing object shapes and specifying object motions. Eg: Cartoon drawing, paintings, product advertisements, logo design

Electronic painting

- Picture is painted electronically on a graphics tablet (digitizer) using a stylus which is cordless and pressure sensitive.

Morphing

- A graphics method in which one object is transformed into another

4. ENTERTAINMENT

Computer graphics methods are used in making motion pictures, music videos, and television shows. The planet and spaceship are drawn in wireframe form and will be shaded with rendering methods to produce solid surfaces. Many TV series regularly employ computer graphics methods. Music videos use graphics in several ways. Graphics objects can be combined with the live action.

Game Industry focus on interactivity, cost effective solutions, avoiding computations and other tricks.

5 EDUCATION AND TRAINING

Computer-generated models of physical, financial, and economic systems are used as educational aids. For some training applications, special systems are designed. Examples of such specialized systems are the simulators for practice sessions or training of ship captains, aircraft pilots, heavy-equipment operators, and air traffic control personnel.

6. VISUALIZATION

Producing graphical representations for scientific, engineering, and medical data sets and processes is referred to as scientific visualization. The term business visualization is used in connection with data sets related to commerce, industry, and other nonscientific areas. There are many different kinds of data sets, and effective visualization schemes depend on the characteristics of the data. Techniques used are color coding, contour plots, graphs, charts, surface renderings & visualizations of volume interiors. Image processing techniques are combined with computer graphics to produce many of the data visualizations.

7. IMAGE PROCESSING

Image processing, applies techniques to modify or interpret existing pictures, such as photographs and TV scans. Two principal applications of image processing are

(1) improving picture quality

(2) machine perception of visual information (robotics)

Image processing and computer graphics are typically combined in many applications. To apply image processing methods

- Digitize a photograph (or picture) into an image file
- Apply digital methods to rearrange picture parts to
 - enhance color separations
 - Improve quality of shading
- Tomography – technique of X-ray photography that allows cross-sectional views of physiological systems to be displayed
- Computed X-ray tomography (CT) and position emission tomography (PET) use projection methods to reconstruct cross sections from digital data

- Computer-Aided Surgery is a medical application technique to model and study physical functions to design artificial limbs and to plan & practice surgery

8. GRAPHICAL USER INTERFACES

It is common now for software packages to provide a graphical interface. A major component of a graphical interface is a window manager that allows a user to display multiple-window areas. Each window can contain a different process that can contain graphical or nongraphical displays. To make a particular window active, we simply click in that window using an interactive pointing device.

Interfaces display menus and icons. Icons are graphical symbols designed to look like the processing option it represents. Advantages of icons are less screen space and easily understood. Menus contain lists of textual descriptions & icons.

Overview of Graphics Systems Video Display Devices

The primary output device in a graphics system is a video monitor. The operation of most video monitors is based on the standard cathode-ray tube (CRT) design. Two types of CRT devices are:

- Refresh CRT
- Color CRT

Refresh Cathode Ray Tube

Basic Operation of CRT

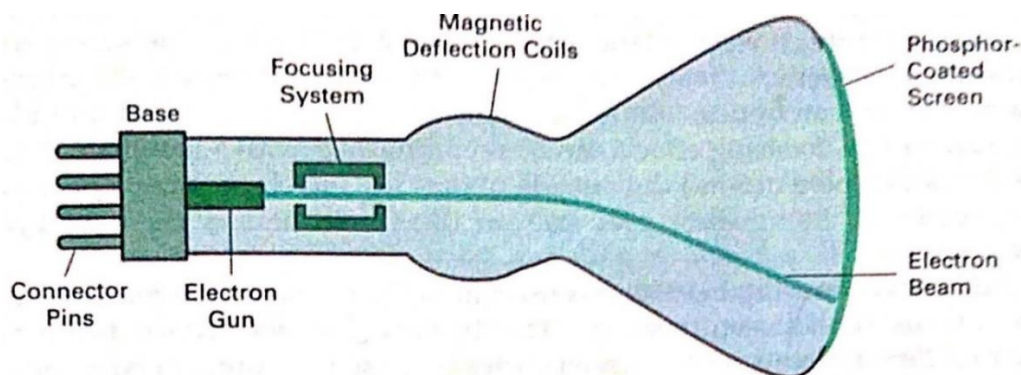


Figure 1: Basic Design of a magnetic-deflection CRT

Figure 1 illustrates the basic operation of a CRT. A beam of electrons (cathode rays), emitted by an electron gun, passes through focusing and deflection systems that direct the beam toward specified positions on the phosphor-coated screen. The phosphor then emits a small spot of light at each position contacted by the electron beam. Because the light emitted by the phosphor fades very rapidly, some method is needed for maintaining the screen picture. One way to keep the phosphor glowing is to redraw the picture repeatedly by quickly directing the electron beam back over the same points (refreshing). This type of display is called a refresh CRT.

The primary components of an electron gun in a CRT are the heated metal cathode and a control grid (Fig. 2) Heat is supplied to the cathode by directing a current through a coil of wire, called the filament, inside the cylindrical cathode structure. This causes electrons to be "boiled off" the hot cathode surface. The free, negatively charged electrons are then accelerated toward the phosphor coating by a high positive voltage. The accelerating voltage can be generated with an accelerating anode.

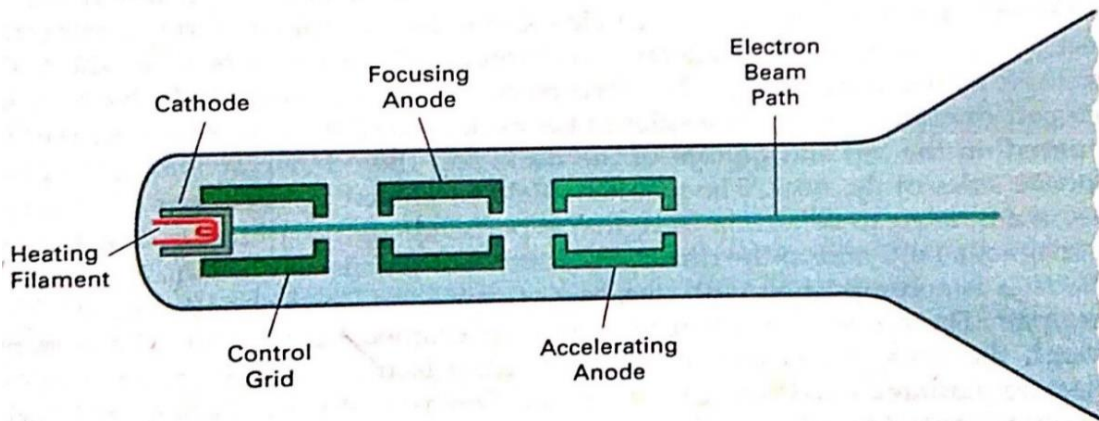


Figure 2.: Operation of an electron gun with an acceleration anode

Intensity of the electron beam is controlled by setting voltage levels on the control grid on the cathode. A smaller negative voltage on the control grid simply decreases the number of electrons passing through. We specify the intensity level for individual screen positions with graphics software commands.

The focusing system in a CRT is needed to force the electron beam to converge into a small spot as it strikes the phosphor. Otherwise, the electrons would repel each other, and the beam would spread out as it approaches the screen. Focusing is accomplished with either electric or magnetic fields.

As with focusing, deflection of the electron beam can be controlled either with electric fields or with magnetic fields. Cathode-ray tubes are now commonly constructed with magnetic deflection coils mounted on the outside of the CRT envelope, as in Fig. 1. Two pairs of coils are used, with the coils in each pair mounted on opposite sides of the neck of the CRT envelope. One pair is mounted on the top and bottom of the neck, and the other pair is mounted on opposite sides of the neck. Horizontal deflection is accomplished with one pair of coils, and vertical deflection by the other pair. The proper deflection amounts are attained by adjusting the current through the coils. (Figure3)

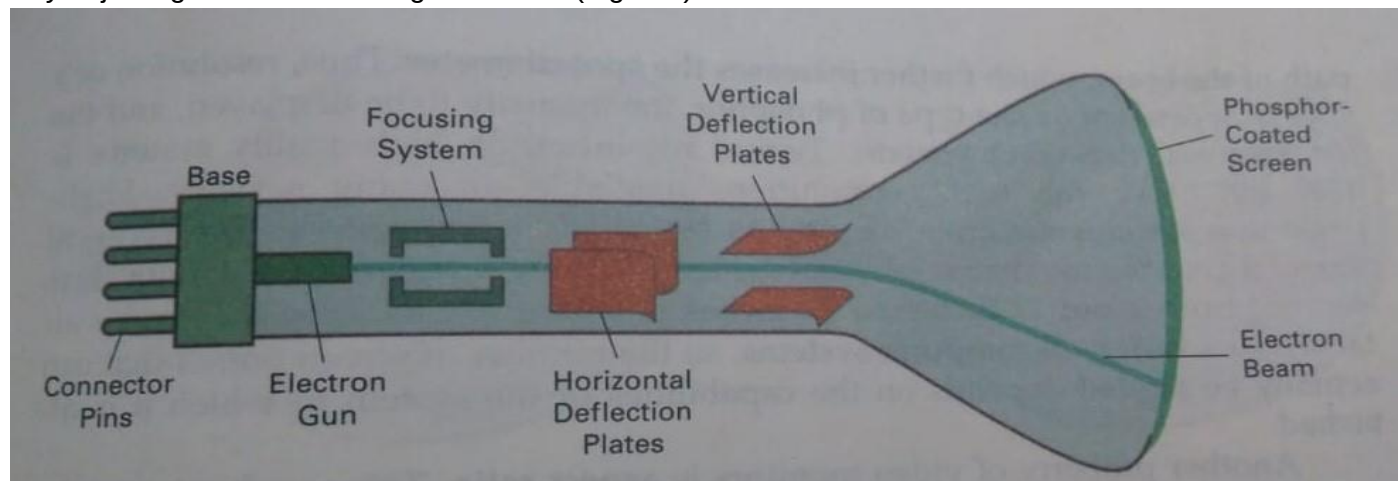


Figure 3: Electrostatic deflection of the electron beam in a CRT

Spots of light are produced on the screen by the transfer of the CRT beam energy to the phosphor. When the electrons in the beam collide with the phosphor coating, they are stopped and their kinetic energy is absorbed by the phosphor.

Persistence

Different kinds of phosphors are available for use in a CRT. Besides color, a major difference between phosphors is their persistence: how long they continue to emit light after the CRT beam is removed.

Persistence is defined as the time it takes the emitted light from the screen to decay to one-tenth of its original intensity. A phosphor with low persistence is useful for animation; a high-persistence phosphor is useful for displaying highly complex, static pictures to emit light.

Resolution

The maximum number of points that can be displayed without overlap on a CRT is referred to as the resolution. Precise definition of resolution is the number of points per centimeter that can be plotted horizontally and vertically. Typical resolution on high-quality systems is 1280 by 1024.

Aspect Ratio

Another property of video monitors is aspect ratio. This number gives the ratio of vertical points to horizontal points necessary to produce equal-length lines in both directions on the screen. (Sometimes aspect ratio is stated in terms of the ratio of horizontal to vertical points.) An aspect ratio of 3/4 means that a vertical line plotted with three points has the same length as a horizontal line plotted with four points.

Refresh Rate

Definition: The number of times per second the image is redrawn.

The entire contents of the frame buffer are displayed on the CRT at a rate high enough to avoid flicker. This rate is called the refresh rate.

For a human to see a steady image on most CRT displays, the same path must be retraced, or refreshed, by the beam at least 60 times per second. Current raster-scan displays perform refreshing at the rate of 60 to 80 frames per second, although some systems now have refresh rates of up to 120 frames per second. Refresh rates are described in units of cycles per second, or Hertz (Hz), where a cycle corresponds to one frame (i.e. a refresh rate of 60 frames per second = 60 Hz).

Raster-Scan Displays

In a raster-scan system, the electron beam is swept across the screen, one row at a time from top to bottom. As the electron beam moves across each row, the beam intensity is turned on and off to create a pattern of illuminated spots. Picture definition is stored in a memory area called the refresh buffer or frame buffer. This memory area holds the set of intensity values for all the screen points. Stored intensity values are then retrieved from the refresh buffer and "painted" on the screen one row (scan line) at a time (Fig. 6). Each screen point is referred to as a pixel or pel (picture element). Home television sets and printers are examples of other systems using raster-scan methods.

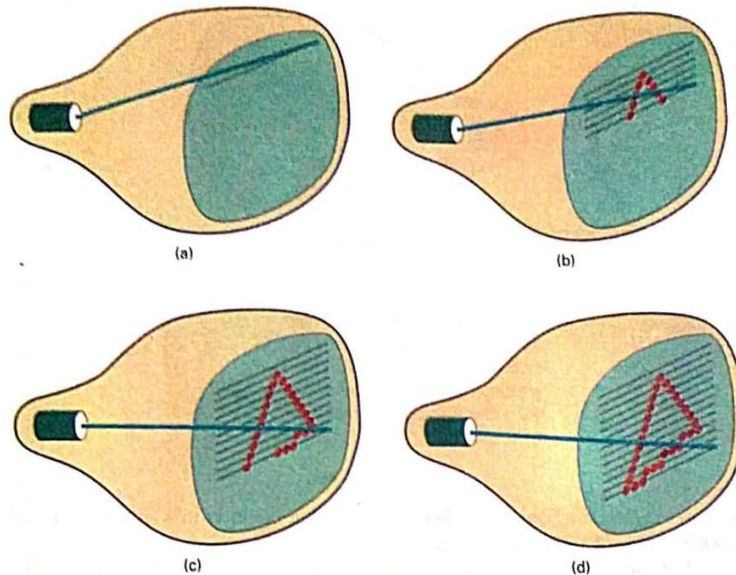


Figure 6: A raster-scan system displays an object as a set of discrete points across each scan line

In a simple black-and-white system, each screen point is either on or off, so only one bit per pixel is needed to control the intensity of screen positions. For a bilevel system, a bit value of 1 indicates that the electron beam is to be turned on at that position, and a value of 0 indicates that the beam intensity is to be off. Additional bits are needed when color and intensity variations can be displayed. Up to 24 bits per pixel are included in high-quality systems. On a black-and-white system with one bit per pixel, the frame buffer is commonly called a bitmap. For systems with multiple bits per pixel, the frame buffer is referred to as a pixmap.

Refreshing on raster-scan displays is carried out at the rate of 60 to 80 frames per second, although some systems are designed for higher refresh rates. At the end of each scan line, the electron beam returns to the left side of the screen to begin displaying the next scan line. The return to the left of the screen, after refreshing each scan line, is called the horizontal retrace of the electron beam. And at the end of each frame (displayed in 1/80th to 1/60th of a second), the electron beam returns (vertical retrace) to the top left corner of the screen to begin the next frame.

On some raster-scan systems (and in TV sets), each frame is displayed in two passes using an interlaced refresh procedure. In the first pass, the beam sweeps across every other scan line from top to bottom. Then after the vertical retrace, the beam sweeps out the remaining scan lines.

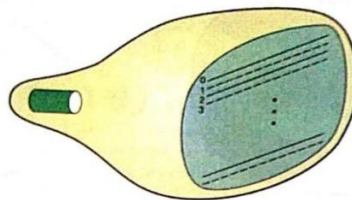


Figure 7: Interlacing Scan lines on a raster-scan display. First , all points on the even-numbered (solid) scan lines are displayed; then all points along the odd-numbered (dashed) lines are displayed.

Interlaced vs. Non-interlaced

➤ Interlaced

- Refresh every other line on each pass
- Lower refresh rate without flicker

- Good for fixed graphics
 - Causes flutter with animated graphics
 - Inexpensive
- Non-interlaced
- Refresh every line on each pass
 - Typical screen sold today

Random-Scan Displays

In random-scan display unit, CRT has the electron beam directed only to the parts of the screen where a picture is to be drawn. Random scan monitors draw a picture one line at a time and for this reason are also referred to as vector displays (or stroke-writing or calligraphic displays). Eg: pen plotter.

Refresh rate on a random-scan system depends on the number of lines to be displayed. Picture definition is now stored as a set of line drawing commands in an area of memory referred to as the refresh display file. Sometimes the refresh display file is called the display list, display program, or the refresh buffer. To display a specified picture, the system cycles through the set of commands in the display file, drawing each component line in turn. After all line drawing commands have been processed, the system cycles back to the first line command in the list.

Random-scan systems are designed for line drawing applications and cannot display realistic shaded scenes.

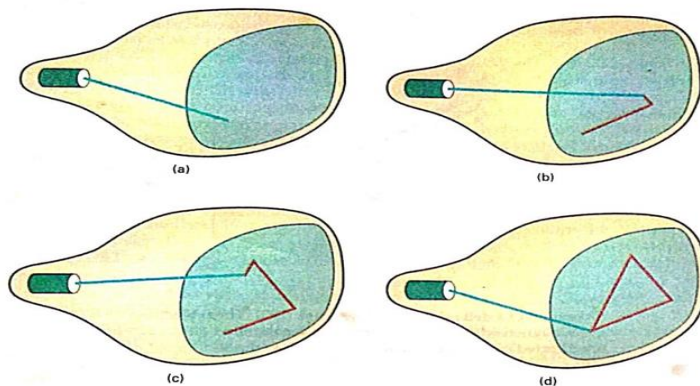


Figure 8: A random-scan system draws the component lines of an object in any order specified.

Bitmap & Pixmap

On a black-and-white system with one bit per pixel, the frame buffer is commonly called a bitmap. For systems with multiple bits per pixel, the frame buffer is referred to as a pixmap

Random Scan	Raster Scan
1. It has high Resolution	1. Its resolution is low.
2. It is more expensive	2. It is less expensive
3. Any modification if needed is easy	3.Modification is tough
4. Solid pattern is tough to fill	4.Solid pattern is easy to fill
5. Refresh rate depends on picture	5. Refresh rate does not depend on the picture.
6. Only screen with view on an area is displayed.	6. Whole screen is scanned.
7. Beam Penetration technology	7. Shadow mask technology
8. It does not use interlacing method.	8. It uses interlacing
9. It is restricted to line drawing applications	9. It is suitable for realistic displays.
10. Pen Plotter	10. TV, Printer

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Color CRT Monitors

To displays color pictures, combination of phosphors is used that emits different-colored light. The two basic techniques for producing color displays with a CRT

- 1.Beam Penetration Method
2. Shadow-mask Method

Beam Penetration Method

The beam-penetration method for displaying color pictures has been used with random-scan monitors. Two layers of phosphor, usually red and green, are coated onto the inside of the CRT screen, and the displayed color depends on are how far the electron beam penetrates into the phosphor layers. A beam of slow electrons excites only the outer red layer. A beam of very fast electrons penetrates through the red layer and excites the inner green layer. At intermediate beam speeds, combinations of red and green light are emitted to show two additional colors, orange and yellow. The speed of the electrons, and hence the screen color at any point, is controlled by the beam-acceleration voltage. Beam penetration has been an inexpensive way to produce color in random-scan monitors, but only four colors are possible, and the quality of pictures is not as good as with other methods.

Shadow-mask Method

Shadow-mask methods are commonly used in raster scan systems (including color TV) because they produce a much wider range of colors than the beam penetration method. A shadow-mask

CRT has three phosphor color dots at each pixel position. One phosphor dot emits a red light, another emits a green light, and the third emits a blue light. This type of CRT has three electron guns, one for each color dot, and a shadow-mask grid just behind the phosphor-coated screen. Figure 9 illustrates the delta-delta shadow-mask method, commonly used in color CRT systems. The three electron beams are deflected and focused as a group onto the shadow mask, which contains a series of holes aligned with the phosphor-dot patterns. When the three beams pass through a hole in the shadow mask, they activate a dot triangle, which appears as a small color spot on the screen. The phosphor dots in the triangles are arranged so that each electron beam can activate only its corresponding color dot when it passes through the shadow mask. In-line arrangement the three electron guns, and the corresponding red-green-blue color dots on the screen, are aligned along one scan line instead of in a triangular pattern. This in-line arrangement of electron guns is easier to keep in alignment and is commonly used in high-resolution color CRT.

We obtain color variations in a shadow-mask CRT by varying the intensity levels of the three electron beams.

Color CRTs in graphics systems are designed as RGB monitors. High-quality raster-graphics systems have 24 bits per pixel in the frame buffer.

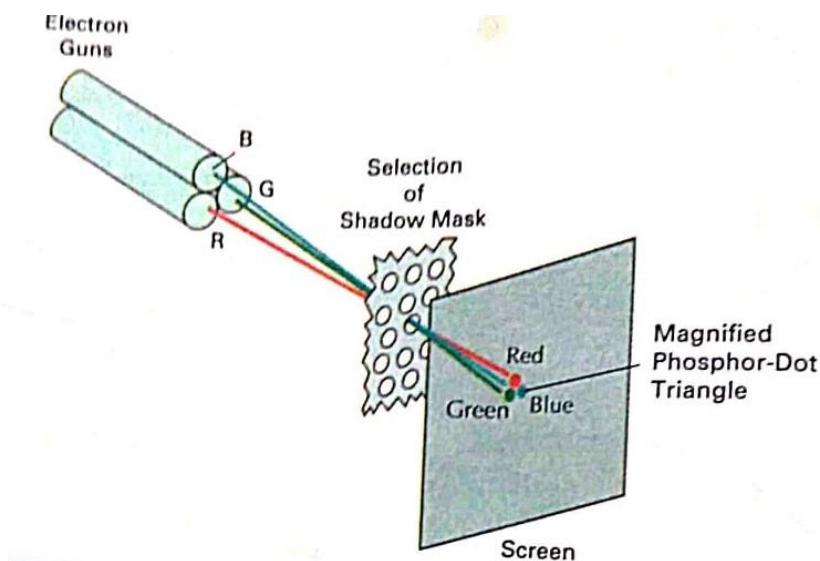
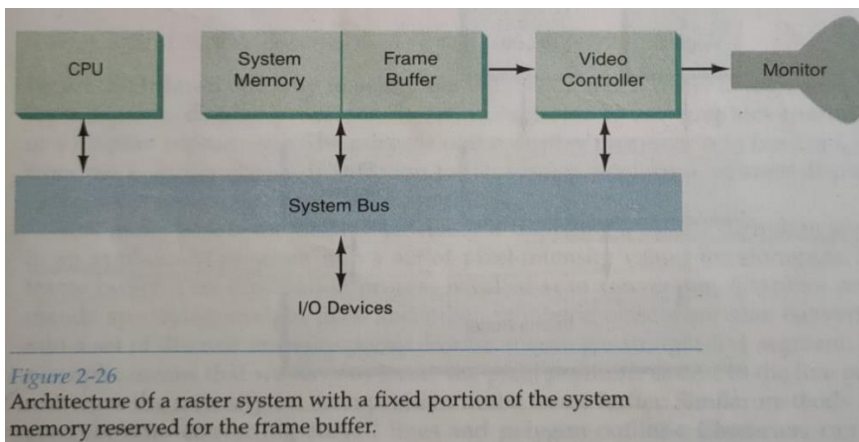


Figure 9: Operation of a delta-delta, shadow-mask CRT. Three electron guns, aligned with the triangular color dot patterns on the screen, are directed to each dot triangle by a shadow mask.

RASTER-SCAN SYSTEMS

Interactive raster graphics systems employ several processing units. In addition to the CPU, a special-purpose processor, called the video controller or display controller, is used to control the operation of the display device.

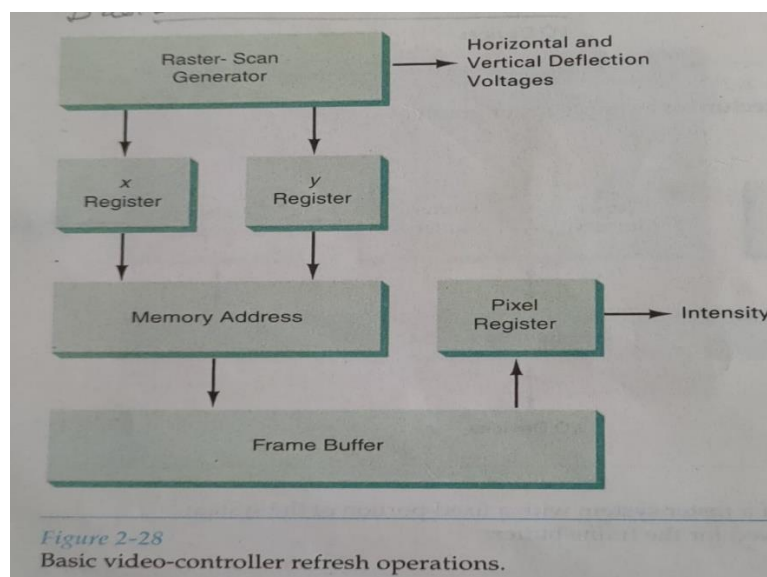
Video Controller



A fixed area of the system memory is reserved for the frame buffer, and the video controller is given direct access to the frame-buffer memory. Frame-buffer locations, and the corresponding screen positions, are referenced in Cartesian coordinates. For many graphics monitors, the coordinate origin is defined at the lower left screen corner. The screen surface is then represented as the first quadrant of a two-dimensional system, with positive x values increasing to the right and positive y values increasing from bottom to top. Scan lines are then labeled from y , at the top of the screen to 0 at the bottom. Along each scan line, screen pixel positions are labeled from 0 to $x_{\max}$.

Basic refresh operations of the video controller(Fig2:28)

Two registers are used to store the coordinates of the screen pixels. Initially, the x register is set to 0 and the y register is set to $y_{\max}$. The value stored in the frame buffer for this pixel position is then retrieved and used to set the intensity of the CRT beam. Then the x register is incremented by 1, and the process repeated for the next pixel on the top scan line. This procedure is repeated for each pixel along the scan line. After the last pixel on the top scan line has been processed, the x register is reset to 0 and the y register is decremented by 1. Pixels along this scan line are then processed in turn, and the procedure is repeated for each successive scan line. After cycling through all pixels along the bottom scan line ($y = 0$), the video controller resets the registers to the first pixel position on the top scan line and the refresh process starts over.



Since the screen must be refreshed at the rate of 60 frames per second, the simple procedure illustrated in Fig. 2-28 cannot be accommodated by typical RAM chips. The cycle time is too slow. To speed up pixel processing, video controllers can retrieve multiple pixel values from the refresh buffer on each pass. The multiple pixel intensities are then stored in a separate register and used to control the CRT beam intensity for a group of adjacent pixels. When that group of pixels has been processed, the next block of pixel values is retrieved from the frame buffer.

The video controller can retrieve pixel intensities from different memory areas on different refresh cycles. In high quality systems, two frame buffers are often provided so that one buffer can be used for refreshing while the other is being filled with intensity values. Then the two buffers can switch roles. This provides a fast mechanism for generating real-time animations.

Raster-Scan Display Processor

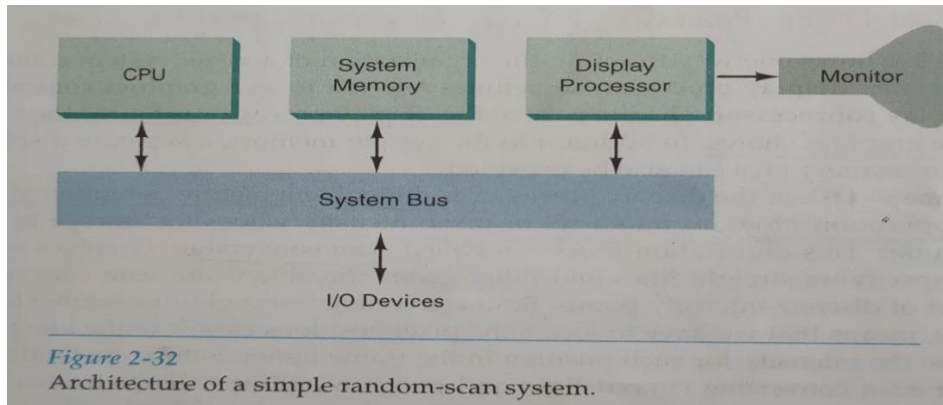
A raster system contains a separate display processor, referred to as a graphics controller or a display coprocessor. The purpose of the display processor is to free the CPU from the graphics chores. In addition to the system memory, a separate display processor memory area can also be provided.

A major task of the display processor is digitizing a picture definition given in an application program into a set of pixel-intensity values for storage in the frame buffer. This digitization process is called scan conversion. Graphics commands specifying straight lines and other geometric objects are scan converted into a set of discrete intensity points. Scan converting a straight-line segment, means that we have to locate the pixel positions closest to the line path and store the intensity for each position in the frame buffer. Similar methods are used for scan converting curved lines and polygon outlines. Characters can be defined with rectangular grids or they can be defined with curved outlines.

Display processors are also designed to perform a number of additional operations. These functions include generating various line styles (dashed, dotted, or solid), displaying color areas, and performing certain transformations and manipulations on displayed objects. Also, display processors are typically designed to interface with interactive input devices, such as a mouse.

In an effort to reduce memory requirements in raster systems, methods have been devised for organizing the frame buffer as a linked list and encoding the intensity information. One way to do this is to store each scan line as a set of integer pairs. One number of each pair indicates an intensity value, and the second number specifies the number of adjacent pixels on the scan line that are to have that intensity. This technique, called run-length encoding, can result in a considerable saving in storage space. A similar approach can be taken when pixel intensities change linearly. Another approach is to encode the raster as a set of rectangular areas (cell encoding). The disadvantages of encoding runs are that intensity changes are difficult to make and storage requirements actually increase as the length of the runs decreases. It is difficult for the display controller to process the raster when many short runs are involved.

RANDOM-SCAN SYSTEMS



An application program is input and stored in the system memory along with a graphics package. Graphics commands in the application program are translated by the graphics package into a display file stored in the system memory. This display file is then accessed by the display processor to refresh the screen. The display processor cycles through each command in the display file program once during every refresh cycle. The display processor in a random-scan system is also referred to as a display processing unit or a graphics controller.

Graphics patterns are drawn on a random-scan system by directing the electron beam along the component lines of the picture. Lines are defined by the values for their coordinate endpoints, and these input coordinate values are converted to x and y deflection voltages. A scene is then drawn one line at a time by positioning the beam to fill in the line between specified endpoints.

Software Standard

The primary goal of standardized graphics software is portability. When packages are designed with standard graphics functions, software can be moved easily from one hardware system to another and used in different implementations and applications. Without standards, programs designed for one hardware system often cannot be transferred to another system without extensive rewriting of the programs.

International and national standards planning organizations have cooperated in an effort to develop a generally accepted standard for computing graphics. This led to the development of the Graphical Kernel System (GKS). A three-dimensional GKS extension was subsequently developed. The second software standard to be developed and approved by the standards organizations was PHIGS (Programmer's Hierarchical Interactive Graphics standard), which is an extension of GKS. Increased capabilities for object modelling, color specifications, surface rendering, and picture manipulations are provided in PHIGS. An extension of PHIGS, called PHIGS+, was developed to provide three-dimensional surface-shading capabilities not available in PHIGS.